Study of the effect of disk and halo parameters on the rotational velocity curve in galaxies dominated by dark matter across all radii.

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Abstract

While considering galaxies like Low Surface Brightness galaxies (LSBs) or dwarf irregular galaxies, the pseudo-isothermal parameters obtained for the dark matter halo happens to be very dominant in contributing to the observed velocity distribution. The disk parameters have negligible influence on the net rotational velocity distribution across all radii. This project is dedicated to the study of the level of influence of the disk parameters (surface brightness and radius parameter) on the rotational velocity curves for the LSBs UGC7321, F568-1, F574-1 and also the Milky way. Goodness-of-fit tests like Kolmogorov Smirnov tests and Chi square tests are performed on the pre-determined disk parameters and fitted halo parameters to determine the freedom with which the parameters could be varied while the modelled rotational velocity curve still remains a good fit to the observed data points. With this exercise we try to establish the fact that the disk does have a significant (though subtle) effect upon the net rotation velocity distribution of the galaxy.

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1.Introduction

Majority of the galaxies in the observable universe are spiral galaxies. Some of the spiral galaxies show very low star formation rate and as a result have low surface brightness. These galaxies are called low surface brightness galaxies. Low surface brightness galaxies are rich in HI gas and the observed extremely high mass to light ratios suggest the dominance of dark matter. The rotational velocity distribution across the radius is determined by measuring the relative red-shifts of the HI emission (21cm) emitted by the interstellar gas in the galaxy. The mass density of the disc is obtained by observing the surface brightness luminosity of the disk. Application of Newtonian mechanics to these observed quantities helps us in understanding the dark matter halo distribution in the galaxies. In this case we use the pseudo isothermal halo model and fit the parameters accordingly.

1.1 Rotation curves of galaxies

It is an observational fact that the visible matter distribution in a spiral galaxy(without a bulge) is in the form of a flat disk whose surface mass density decreases exponentially along the radial direction. The surface density is expressed in the following form for an exponential disk given in units of solar masses per square kiloparsecs:

$$\Sigma(R) = \Sigma_0 \, e^{\frac{-R}{R_D}} \tag{1}$$

Where Σ_0 is the central surface density and R_D is the scale factor for the disk radius.

In order to theoretically obtain the rotation velocity distribution for the spiral galaxy, we need to find the potential due to the disk as well as the pseudo isothermal halo surrounding the disk. And accordingly calculate the rotation velocity from the expression:

$$v^2 = R \frac{d\Phi}{dR} \tag{2}$$

The resultant rotational velocity of the galaxy is the quadrature sum of disk rotation velocity as well as that of the halo.

1.1.1 Obtaining the potential for an exponential disk via Bessel functions:

Considering the spiral galaxy to be a flat disk in the cylindrical polar coordinates with the z-axis passing through the center perpendicular to the plane of the disk, we first solve for the Laplace's equation $\nabla^2 \Phi = 0$ (This is the Toomre's method as given in Toomre 1962). Applying the boundary conditions on the disk and at infinity we obtain the following expression for the potential of the disk at z = 0 (Binney&Tremaine 1987).

$$\Phi(R, 0) = -\pi G \Sigma_0 R [I_0(y) K_1(y) - I_1(y) K_0(y)], \quad \text{where } y = R/R_D \tag{3}$$

Where $I_0(y)$, $K_1(y)$, $I_1(y)$, $K_0(y)$ are special Bessel functions.

Solving the equation (2) by substituting (3) we obtain the expression for the rotational velocity to be :

$$v_c^2(R) = 4 \pi G \Sigma_0 R_D y^2 [I_0(y) K_0(y) - I_1(y) K_1(y)], \quad \text{where } y = R/R_D \tag{4}$$

Considering the Milky-way galaxy, the disk parameters as per observations are $\Sigma_0 = 700\ M_\odot\ \text{pc}^{-2}$ and $R_D = 3.2$ kpc. With these the rotation velocity curve as a function of radius looks like the following:

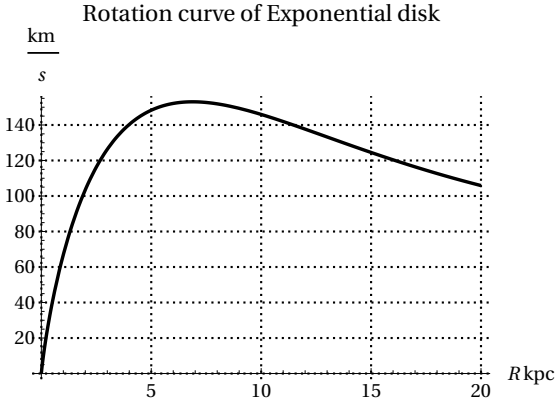


Figure 1. This is the plot of rotation velocity versus the radius for the disk in Milky way galaxy

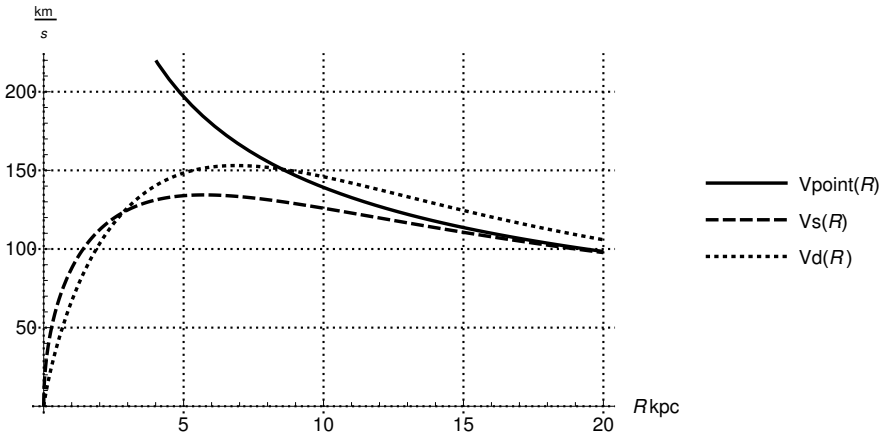


Figure 2. Plot showing the rotational velocity of an exponential disk (Vd) , sphere with the same mass as that of the disk (Vs), Point source with the same total mass as that of the disk (Vpoint)

1.1.2 Pseudo-isothermal halo:

We consider a pseudo-isothermal halo whose density distribution is spherically symmetric in order to account for the missing mass in the galaxy and explain the observed rotation curve. The density expressed as a function of radius is as follows:

$$\rho(R) = \frac{\rho 0}{1 + \frac{R^2}{\text{Rc}^2}} \tag{5}$$

Where $\rho 0$ is the central halo density and Rc is the core radius of the halo.

Solving for the potential for this distribution using Poisson's equation , we obtain the following expression (Ghosh&Jog 2014) :

$$\phi_{\text{halo}}(R,\, z) \;=\; \left(4\ \text{Pi}\ G\,\rho 0\ \text{Rc}^2\right)\left(0.5 * \text{Log}\left[\text{Rc}^2 + R^2 + z^2\right] + \frac{\text{Rc}}{\sqrt{R^2 + z^2}} * \text{ArcTan}\left[\frac{\sqrt{R^2 + z^2}}{\text{Rc}}\right] - 1\right) \tag{6}$$

This leads to the expression for the rotational velocity (at z = 0) :

$$v_{\text{halo}}^2(R) \;=\; \left(4\ G\,\text{Pi}\ R\,\text{Rc}^2\ \rho 0\right)\left(\frac{1}{R\left(1 + \frac{R^2}{\text{Rc}^2}\right)} + \frac{1.\ R}{R^2 + \text{Rc}^2} - \frac{R\,\text{Rc}\,\text{ArcTan}\left[\frac{\sqrt{R^2}}{\text{Rc}}\right]}{\left(R^2\right)^{3/2}}\right) \tag{7}$$

1.1.3 Net rotational velocity

The net rotational velocity is obtained by the following expression:

$$v_{\text{net}}{}^2(R) \;=\; v_c^2(R) \;+\; v_{\text{halo}}^2(R) \tag{8}$$

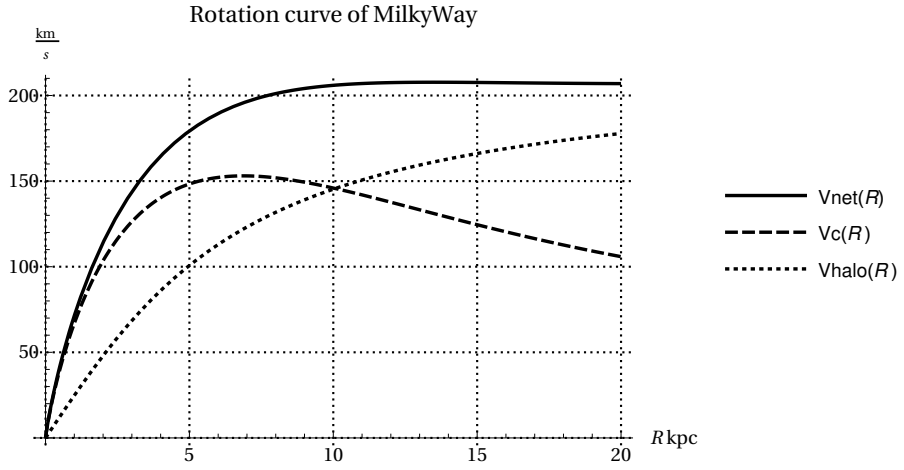


Figure 3. Rotation curve of the halo in the Milky way galaxy

Table 1. Input parameters for different galaxies

Galaxy	Σ_0 $M_\odot\ \text{pc}^{-2}$	R_D (kpc)	ρ_0 $M_\odot\ \text{pc}^{-3}$	R_c (kpc)
UGC7321[1]	50.2	2.1	0.126	1.4
F568 – 1[2]	27.98	5.3	0.081	2.32
F574 – 1[2]	44.35	4.3	0.056	1.82

[1] Σ_0 and R_D were obtained from Banerjee et al. (2010) while ρ_0 and R_c were obtained by fitting the isothermal halo to the observed data points

[2] Σ_0 and R_D were obtained from de Blok et al. (2001) AJ. while ρ_0 and R_c were obtained by fitting the isothermal halo to the observed data points

2. Analysis of the LSB galaxy UGC7321

UGC7321 is a bulge-less superthin galaxy with a dense compact halo. The input parameters are mentioned in Table-1. The values for Σ_0 and R_D were obtained from Banerjee et al.(2010) while that of ρ_0 and R_c were obtained by fitting the curve to the observed data points due to inconsistencies with the input parameters mentioned in Banerjee et al.(2010) with the observed data points.

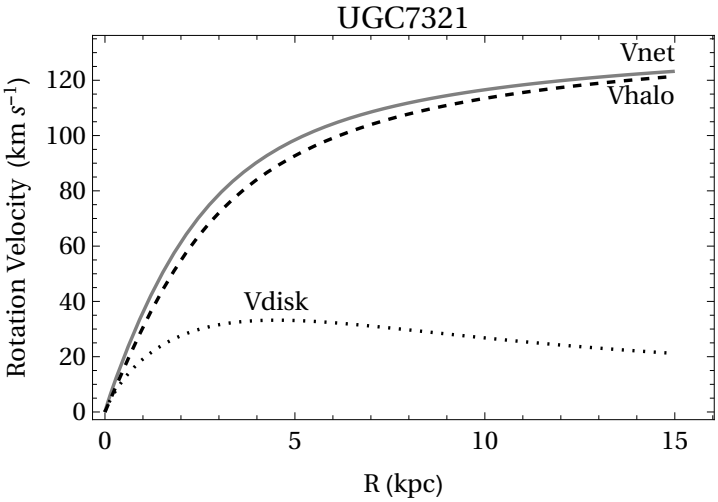


Figure 4. Theoretical Rotational velocity curves for disk (Vc), halo (Vhalo) and disk + halo (Vnet).

The observed rotation velocity values for UGC7321 were available across the radii with the net recession velocity of the galaxy being 406.800 km/s as observed from the redshift at the center of the edge-on galaxy. Subtracting this value from the observed velocities across all radii, we obtain the rotational velocity distribution.

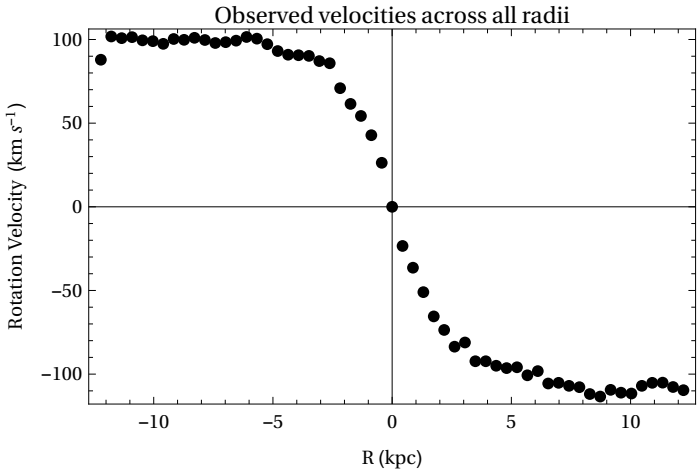


Figure 5. Observed data points for UGC7321 after subtracting the recessional velocity.

We take the average of the modulus of the rotational velocities in the north and south direction for the same radii and obtain the generalized distribution of the velocity along the radius which could point along any direction from the center in the z = 0 plane. Using these values we fit the theoretical expression for the net rotational velocity in order to obtain the halo parameters knowing the disk parameters from observations.

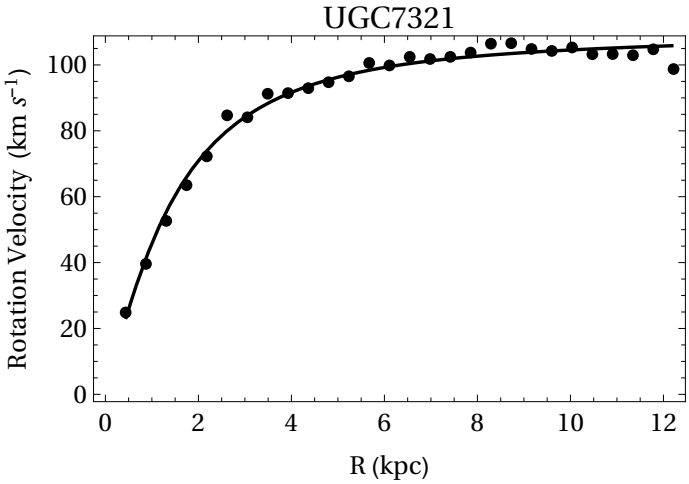


Figure 6. Observed data points and the fitted curve for UGC7321

2.1 Playing around with the disk parameters:

As can be noticed in Figure 4 , the net rotation velocity of UGC7321 is very nearly the same as that of the halo(alone). With this we can infer that the disk mass plays a negligible role in determining the rotation velocities of the galaxy. Hence we make an attempt in determining the freedom with which the disk parameters can be varied such that the net rotation curve still remains a good fit to the observed data points.

2.1.1 Kolmogorov Smirnov test

Kolmogorov Smirnov test (KS test) is a goodness-of-fit test where the consistency between the observed data and the theoretical prediction of the distribution is being tested. Among the various tests available for goodness-of-fit testing, KS test turns out to be the most sensitive test even for the bivariate data that we have used as justified in Peacock(1983). KS test is the simplest yet most efficient method when it comes to testing the level of misfit between the data and the expected curve. It considers the largest absolute differences between the two cumulative distributions as a measure of misfit. The Kolmogorov–Smirnov test effectively uses a test statistic based on $\mathbf{sup}_x \left| \hat{F}(x) - F(x) \right|$ where $\hat{F}(x)$ is the empirical CDF of the observed data and $F(x)$ is the CDF of distribution. In this case we have bivariate data and its p-value is obtained by summing the univariate marginal p-values (where p-value is the probability with which the data points could have been extracted from the given distribution). The central surface density Σ_0 for UGC7321 is 50.2 $M_\odot$ pc^{-2} and the disk parameter R_D is 2.1 kpc. As seen earlier, these parameters have little effect on the net rotational velocity of the LSB galaxy. Hence slight variations in the values of these parameters might not affect the resultant curve as it is mainly dependent on the halo parameters. We perform KS tests to an accuracy of 95%, 98% and 99% on the curve with the observed data by varying the disk parameters with Σ_0 varying from 0 to 200.8$M_\odot$ pc^{-2} in steps of 4.79 $M_\odot$ pc^{-2} and R_c varying from 1.29 kpc to 2.91 kpc in steps of 0.032 kpc. For each value of Σ_0 and R_D KS test is performed and the following results were obtained:

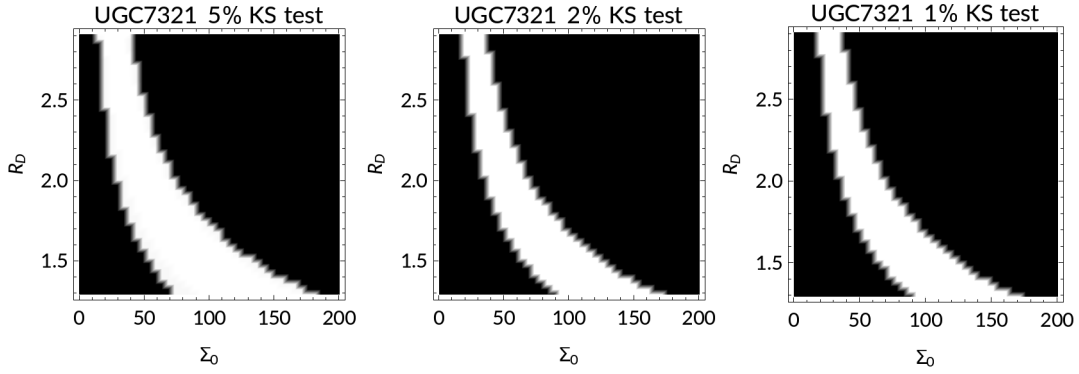


Figure 7. KS tests for UGC7321

As seen in the results we can infer that the disk mass cannot be completely neglected as the range around $\{\Sigma_0, R_D\} = \{0,0\}$ is not considered a good fit from the KS test to an accuracy of 95% (same is the case with 98%and 99%). It is obvious to note that the area of the allowed parameter space decreases with increase in accuracy. Even with 99% accuracy we can see that there is a wide range of allowed values for $\Sigma_0$and R_D. The observed values for $\Sigma_0$and $R_D$might not be accurate and different detectors might yield different parameters bringing up various contradictions while constructing the models for the LSBs. One example can be found in Arunima&Bapat 2017 where the change in disk parameters due to the extraction of data from a different source led to large changes in the

halo parameters from that previously obtained in Arunima&Jog 2010.

2.1.2 Chi squared test

Chi Square test being the standard test for curve fitting also provides hints in discussing the properties of the disk parameters. Pearson's Chi square test is being performed. The Pearson χ^2 test effectively compares a histogram of data to a theoretical histogram based on the distribution. The bins are chosen to have equal probability in the distribution. For univariate data, the test statistic is given by $\sum_{i=1}^{n} (o_i - e_i)^2 / e_i$, where o_i and e_i are the observed and expected counts for the ith histogram bin, respectively. In our case of testing bivariate data, sum of univariate marginal p-values is used . This test is being performed to show the differences between chi-square test and KS test. As is evident from figure 8., chi-square test is not as sensitive and not very reliable in

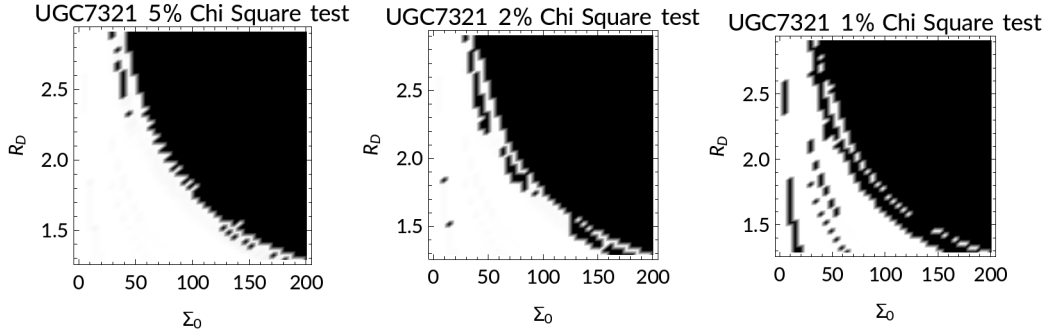
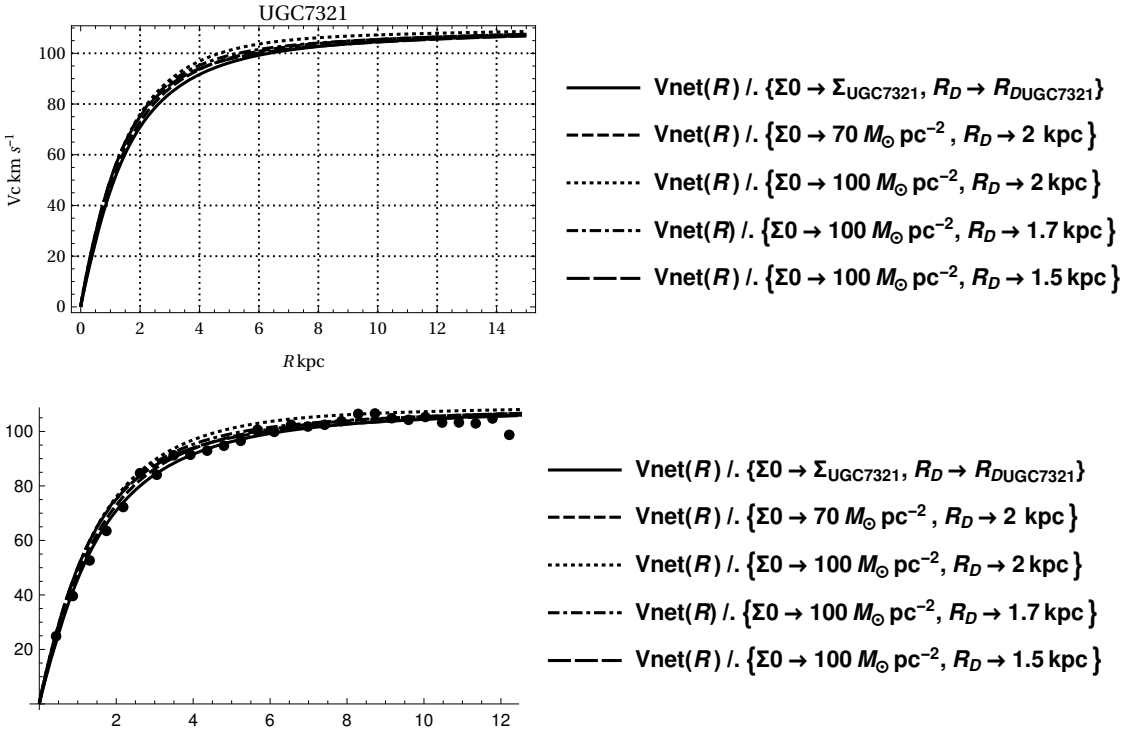


Figure 8. Chi Square tests for UGC7321



3. Analysis of F568-1

F568-1 is another LSB and the data is being taken from de Blok et al. (AJ, 2001). This galaxy along with the galaxy F574-1 discussed in §4 are specifically chosen out of the given 30 high-resolution mass models in de Blok et al. (AJ, 2001) because the data points for these galaxies more or less lie on the fitted curve and there is not much deviation from the curve. Hence KS tests would yield fruitful results. In case the data points had large deviations from the fitted curve (large χ^2 value) then the errors associated with these data points would overshadow the difference which the variance in the disk parameters would make on the fitted curve. Hence these two galaxies were judiciously chosen in order to smoothly carry out the goodness-of-fit tests and accordingly draw various conclusions which is discussed in the later sections. The halo parameters taken for the minimum disk as given in the paper gives $\rho_0 = 0.097$ $M_\odot$ pc^{-3} and $R_c = 2.22$ kpc. Due to inconsistencies with the observed data points, fitting the curve gives different values for ρ_0 to be

$0.081 M_\odot$ pc^{-3} and R_c to be 2.32 kpc. The disk parameters given in de Blok et al. are μ(B) = 23.8 mag arcsec^{-2} and R_D = 2.8 kpc. The given surface brightness is converted to the surface mass density using the following method(Binney & Merrifield 1998) :

$$\mu(R, z) = -2.5 \log\left(I / L_\odot \text{ pc}^{-2}\right) + M_\odot + 21.572 \tag{9}$$

For B-band $M_\odot$ = 5.48 (Cox 2000.)

From (9) we obtain the surface brightness to be 19.99 $L_\odot$ pc^{-2}. The surface density is given by $M_\odot / L_\odot$ times the intensity, where the value of $M_\odot / L_\odot$ for B-band is 1.4. Hence the calculated surface density for F568-1 is 27.98 $M_\odot$ pc^{-2}.

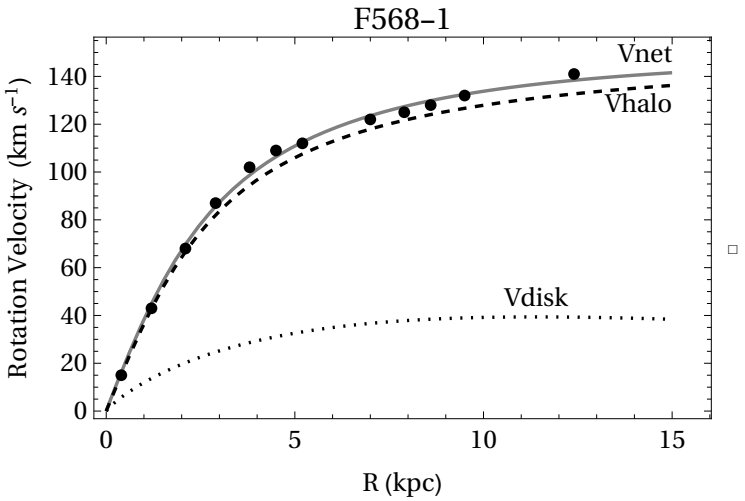


Figure 9. Rotational velocity curves for disk (Vc), halo (Vhalo) and disk + halo (Vnet) along with the observed data points for F568-1

3.1.1 Kolmogorov Smirnov test

Similar to § 2.1.1 KS test was being performed on the disk parameters of the galaxy F568-1 with Σ_0 varying from 0 to $112 M_\odot$ pc^{-2} in steps of 4.79 $M_\odot$ pc^{-2} and R_c varying from 4.5 kpc to 6.2 kpc in steps of 0.032 kpc. The surface density being low for this galaxy when compared to that of UGC7321, disk mass becomes prominent in determining the net rotational velocity after Σ_0 = 46 $M_\odot$ pc^{-2}. Low surface density also allows one to completely ignore the disk mass as the regions around $\{\Sigma_0, R_c\} = \{0,0\}$ are allowed even in KS test performed with the accuracy of 99%.

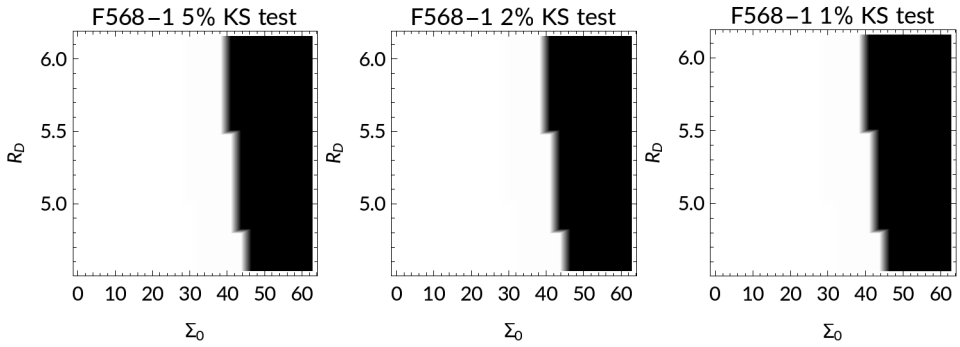


Figure 10. KS tests for F568-1

3.1.2 Chi square test

Chi square test provides a similar result as that of the KS test in the case of the LSB F568-1.

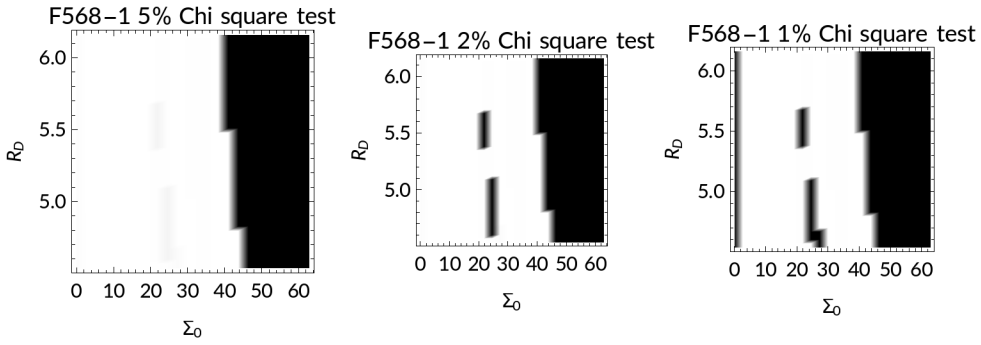


Figure 11. Chi Square tests for F568-1

4. Analysis of F574-1

F574-1 is an LSB taken from de Blok et al. (AJ, 2001). With the known surface brightness, the surface density was being calculated by using (9) to obtain the value of the surface density to be 44.35 $M_\odot$ pc^{-2}. Again due to inconsistencies with the given halo parameters, the halo parameters are independently obtained by fitting the curve with the observed data points with the known disk parameters. These halo parameters are approximate values and are sufficient for the tests being performed in this project. But it is important to remember that these values cannot be relied upon as accurate as the data points were taken into consideration by directly reading out the values from the plots given in de Blok et al. (AJ, 2001) which would be possessing human errors at each data point. Same is the case for F568-1.

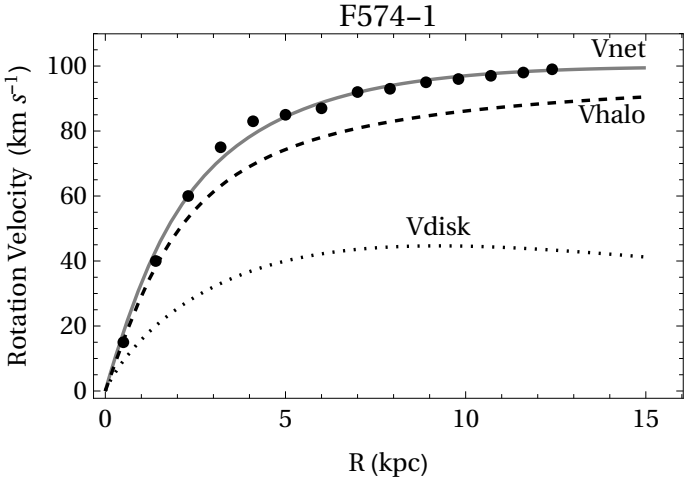


Figure 12. Rotational velocity curves for disk (Vc), halo (Vhalo) and disk + halo (Vnet) along with the observed data points for F574-1

4.1.1 Kolmogorov Smirnov test

In the case of F574-1, the comparatively large values of disk parameters increases the disk contribution in the net rotation velocity curve. Hence KS test results in a smaller region of allowed parameters. The test is performed with Σ_0 varying from 0 to 177$M_\odot$ pc^{-2} in steps of 4.79 $M_\odot$ pc^{-2} and R_c varying from 3.2 kpc to 4.86 kpc in steps of 0.032 kpc.

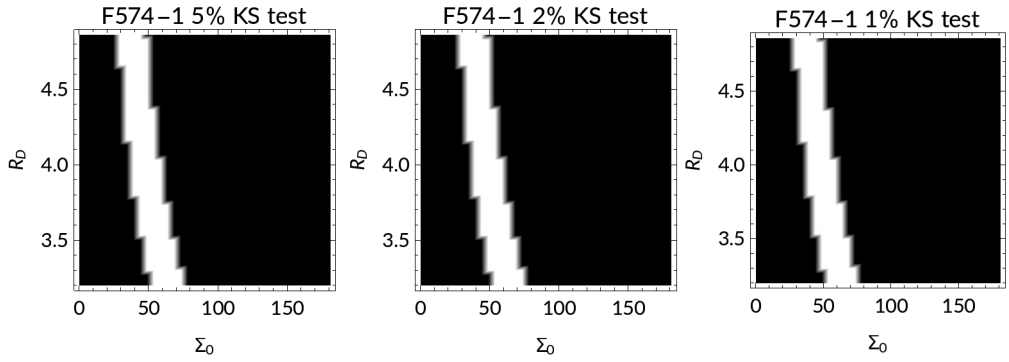


Figure 13. KS tests for F574-1

4.1.2 Chi square test

Here we can notice a similar case as that of UGC7321.

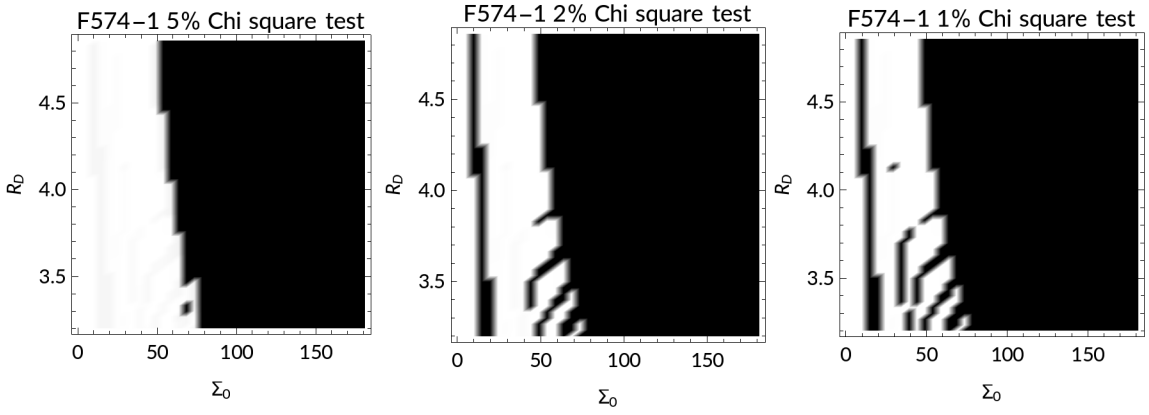


Figure 14. Chi Square tests for F574-1

Discussions

As seen in the above analyses, the range of allowed disk parameters happens to be widely ranged. A similar procedure followed for the Milky-way galaxy yields not result i.e., there appears no significant parameter space allowed to be manipulated freely. This firmly establishes the fact that the disk contribution in the Milky way is very significant when compared to LSBs.

Conclusions

With these analyses performed, inferences can be drawn about the level of significance of the disk parameters. The disk is expected to have a minimum surface density and a reasonable radius parameter in order to show the widely observed rotational velocity distributions. This could throw some light in understanding the galaxy formation processes wherein the minimum requirement for a galaxy to have a reasonable rotational velocity distribution can be approximately established.

Acknowledgments

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